Sr.No.9836

Master of Computer Application (MCA) (Two Yrs Programme) - 3rd Semester (Batch 2024-26) (CBEGS)

(2225)

Paper: CSL-5042 : Advanced Computer Architecture

Time allowed: 3 hrs. **Max. Marks: 70**

NOTE: Students are required to attempt five questions, selecting at least one question from each Section. The fifth question may be attempted from any Section.

SECTION A

Q1. (a) Draw the control gates circuit diagram associated with the Address Register (AR). Use the following control signals to design the circuit: [7]

$$R'T_0: \ AR \leftarrow PC$$
$$R'T_2: \ AR \leftarrow IR(0-11)$$
$$D_7IT_3: \ AR \leftarrow M[AR]$$
$$RT_0: \ AR \leftarrow 0$$
$$D_5T_4: \ AR \leftarrow AR+1$$

(b) A digital computer has a memory unit of 64 KB and requires 8-bit words. The system uses RAM chips of size 8K × 8 and ROM chips of size 4K × 8. Determine the number of RAM and ROM chips required to implement the memory unit. Show all necessary calculations and draw a neat circuit diagram illustrating the memory organization. [7]

Q2. (a) A computer system has a main memory of 128 KB and a cache of 8 KB with a block size of 32 bytes. Calculate the following for the cache in (i) direct-mapped and (ii) 4-way set-associative organizations. Show all working steps.

(i) Number of tag bits required (for direct-mapped and for 4-way set-associative).

(ii) Number of blocks in the cache (direct-mapped).

(iii) Number of sets in the cache (4-way set-associative).

(iv) Does the number of block-offset bits change when the mapping is changed from direct-mapped to 4-way set-associative? Explain and justify your answer. [7]

(b) Explain the circuit diagram of a Hardwired Control Unit and describe how it generates the control signals required for executing an instruction. [7]

SECTION B

Q3. The following reservation table shows the usage of three pipeline stages S_1, S_2, and S_3 during the execution of a single operation. Each "X" represents the clock cycles during which the corresponding stage is reserved. Determine the minimum average latency for a sequence of operations, showing each step clearly. [14]

Time →

	1	2	3	4	5	6	7	8
S_1	X					X		X
S_2		X		X				
S_3			X		X		X	

Contd...P/2

Q4. (a) What are hazards in instruction pipelining? Explain the various types of data hazards, distinguishing between true and false dependencies, and discuss the techniques used to handle them. [7]

(b) Consider a 5-stage instruction pipeline without branch prediction: Fetch (F), Decode (D), Operand Fetch (OF), Execution (E), and Write Back (WB), with stage delays of 5 ns, 7 ns, 10 ns, 8 ns, and 6 ns, respectively. Each intermediate buffer adds a 1 ns delay. A program of 10 instructions (I_1–I_{10}) executes on this processor, where I_4 is a branch with target I_8. The branch is taken, and its outcome is known at the end of the E stage. Find the total execution time (in ns) for the program and draw the corresponding time–stage diagram of the pipeline. [7]

SECTION C

Q5. Write short notes on the following:

(a) Superscalar Pipeline Design [7]

(b) Superpipelined Design [7]

Q6. Explain the architecture of SIMD array processors. Describe how masking and data routing mechanisms work in SIMD array processors with the help of a neat diagram. [14]

SECTION D

Q7. Explain the Static Connection Network and Dynamic Connection Networks used in multiprocessor systems. Discuss their structure, operation, and provide suitable examples with diagrams. [14]

Q8. (a) Differentiate between tightly coupled and loosely coupled multiprocessor systems. [7]

(b) Explain interconnection structures used in multiprocessor systems. Consider a crossbar switch that connects four processors (P1–P4) to four memory modules (M1–M4). Draw a neat diagram showing all crosspoints as switches and indicate one active path between P2 and M3. If each crosspoint costs Rs.10, calculate the total number of switches and the total hardware cost. [7]

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